

SANMATI RAJIV SAWALWADE

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SUMMARY

Product-minded engineer who turns data, experimentation, and analysis into product decisions, with a focus on search relevance, growth, and business analysis (Python, SQL). Strong in A/B experimentation, hypothesis testing, and predictive modeling, with a track record of translating findings into product and business decisions for technical and non-technical stakeholders. Three years shipping data and ML products to 10M+ users, plus a full job-search product built solo from concept to live. Detail-oriented, learns fast, and owns work 0-to-1.

EDUCATION

Northeastern University - M.S. Information Systems

Expected May 2027

GPA: 4.0 / 4.0 | Coursework: Machine Learning, Data Science Engineering, Business Analysis, Statistics

Dr. D. Y. Patil College of Engineering - Bachelor of Engineering

Mar 2022

SKILLS

Statistics & Modeling: Bayesian methods, hypothesis testing, A/B experimentation, survival analysis, calibration, cross-validation; predictive modeling (lifetime value, churn, conversion), regression/classification, ensemble methods

Languages & Libraries: Python (advanced; pandas, NumPy, scikit-learn, statsmodels, TensorFlow, PyTorch), SQL (JOINS, GROUP BY, window functions, CTEs, query optimization), R; Java, C/C++

Data & Tools: BigQuery, Airflow, Spark/PySpark, ETL, feature engineering; Tableau, Power BI, plotly/matplotlib/seaborn; FastAPI, PostgreSQL, Docker, Git, Linux CLI, cloud deployment

AI: Claude API, AI-assisted development, prompt design and verification; RAG / vector search (Pinecone), LangChain

EXPERIENCE

Audifytech - Software Engineer

Sep 2022 - Dec 2023

- Designed and ran a Bayesian A/B experimentation framework across 15+ experiments with calibrated readouts, translating results into product decisions that lifted D7 retention 7% and protected \$140K+ ARR
- Built production predictive models: a clustering pipeline segmenting 100K+ users (K-Means + DBSCAN) and a retrieval layer powering a recommender with a 23% precision lift
- Shipped a production classification model (0.82 AUC-ROC) serving 1M+ users; built monitoring (drift tracking, calibration checks, latency profiling) and validated results against business outcomes

Gameskraft (\$2.3B unicorn) - Software Engineer Intern

Sep 2021 - Dec 2021

- Built ensemble models on a 10M+ event dataset to estimate customer lifetime value and surface its key drivers, reducing first-week churn 12% across products serving 10M+ users
- Owned statistical validation across 15+ production releases (hypothesis testing, survival analysis) to a zero-regression bar, eliciting hypotheses from product and engineering partners and testing them against data

Pocket52 - Software Engineer Intern

Nov 2020 - Aug 2021

- Ran exploratory analysis and modeling in SQL/Python on a product serving 10M+ users, investigating conversion and retention with a hypothesis-driven approach and building dashboards stakeholders relied on for decisions

PROJECTS

FirstApply | AI Job-Search Product, solo-built concept to live ([Link](#))

Sep 2025 - Present

- Built and shipped a job-search product (Python, Node.js, FastAPI, PostgreSQL, Docker): real-time ingestion crawling 6+ sources via Playwright + BeautifulSoup, aggregating 2,600+ records with automated deduplication and structured relational storage
- Built an NLP relevance-ranking engine on the Claude API scoring each record 0-100 against up to 5 user-intent profiles, surfacing top matches via a near-zero-latency alerting pipeline; designed and verified the prompts and model output
- Owned the product 0-to-1: data model, ranking logic, API design, deployment, and iteration - end-to-end, AI-assisted building in a fast, ambiguous environment

Semantic Search & Document Retrieval System

- Designed a multi-stage semantic search pipeline (Transformer embeddings, semantic chunking, Pinecone vector search) ranking results for sub-200ms retrieval over 50K+ documents at 89% answer relevance; deployed via FastAPI + Docker with LangChain orchestration

COVID-19 CT Scan Classifier | Medical image classification ([Link](#))

- Built a deep-learning classifier triaging chest CT scans (COVID-19 / Pneumonia / Normal); fine-tuned EfficientNetB0 via two-stage transfer learning in TensorFlow/Keras, reaching 93.4% test accuracy (0.91-0.97 per-class F1), validated with 5-fold stratified cross-validation

Meshy Prompt-to-Success | Experimentation case study, text-to-3D quality ([Link](#))

- Ran 6 controlled experiments across object types using objective topology metrics; quantified quality deltas and published findings as a full case study - structured experimentation end-to-end in Python on a \$0 stack